T2 — Kinetics, Equilibrium and Redox

2.1 Rates of Reaction

Rate of reaction: The change in concentration of a reactant or product per unit time.
Units: mol dm^{-3} s^{-1}.

Rate = -d[reactant]/dt = +d[product]/dt

Rate can be measured experimentally by monitoring changes in:

- **Volume of gas produced** (gas syringe or inverted burette) — suitable when a gaseous product is formed
- **Mass loss** (balance) — when a gaseous product escapes
- **Colour intensity** (colorimetry/spectrophotometry) — when a coloured reactant or product is involved
- **Electrical conductivity** — when the number or type of ions changes
- **pH change** — when H^+ or OH^- is consumed or produced
- **Clock reactions** (initial rate method) — time for a visual change to occur

The **instantaneous rate** at any point can be found from the gradient of the tangent to a concentration-time graph at that point. The **initial rate** is the gradient at t = 0.

Factors Affecting Rate

Factor	Effect	Explanation (using collision theory)
Concentration	Higher concentration increases rate	More particles per unit volume, so more frequent collisions

Temperature	Higher temperature increases rate significantly	Particles have greater kinetic energy and move faster → (1) more frequent collisions, and more importantly (2) a much greater proportion of collisions exceed E_a
Surface area	Greater surface area increases rate	More surface exposed for reactant particles to collide with (e.g. powder vs lumps)
Catalyst	Increases rate	Provides an alternative reaction pathway with a lower activation energy
Pressure (gases)	Higher pressure increases rate	Gas molecules forced closer together, increasing collision frequency

2.2 Collision Theory

Collision theory: For a reaction to occur, reactant particles must: (1) **collide** with each other, (2) with **sufficient energy** (equal to or greater than the activation energy, E_a), and (3) with the **correct orientation** so that bonds can break and reform.

Not all collisions lead to a reaction — only those that meet both the energy and orientation requirements are termed **successful collisions**. Increasing temperature has a dual effect: it increases collision frequency slightly (particles move faster), but more importantly, it dramatically increases the proportion of molecules with energy ≥ E_a.

A useful approximation: For many reactions near room temperature, a 10 K rise in temperature approximately doubles the rate. This is because the fraction of molecules with energy ≥ E_a increases exponentially with temperature (described mathematically by the Arrhenius equation).

2.3 Maxwell-Boltzmann Distribution

The Maxwell-Boltzmann distribution shows the spread of kinetic energies (or speeds) among molecules in a gas sample at a given temperature. Key features of the curve:

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- The curve starts at the **origin** (no molecules have zero kinetic energy at any temperature above 0 K, but the probability density at exactly zero energy is zero)
- The **peak** represents the most probable energy (E_{mp}) — the energy possessed by the largest number of molecules
- The curve is **asymmetric** with a long tail extending to the right (a small number of molecules have very high energies)
- The curve **never touches the x-axis** at high energies (it approaches it asymptotically)
- The **total area** under the curve represents the total number of molecules and is constant
- The **shaded area** to the right of E_a represents molecules with sufficient energy to react

Effect of Temperature on the Distribution

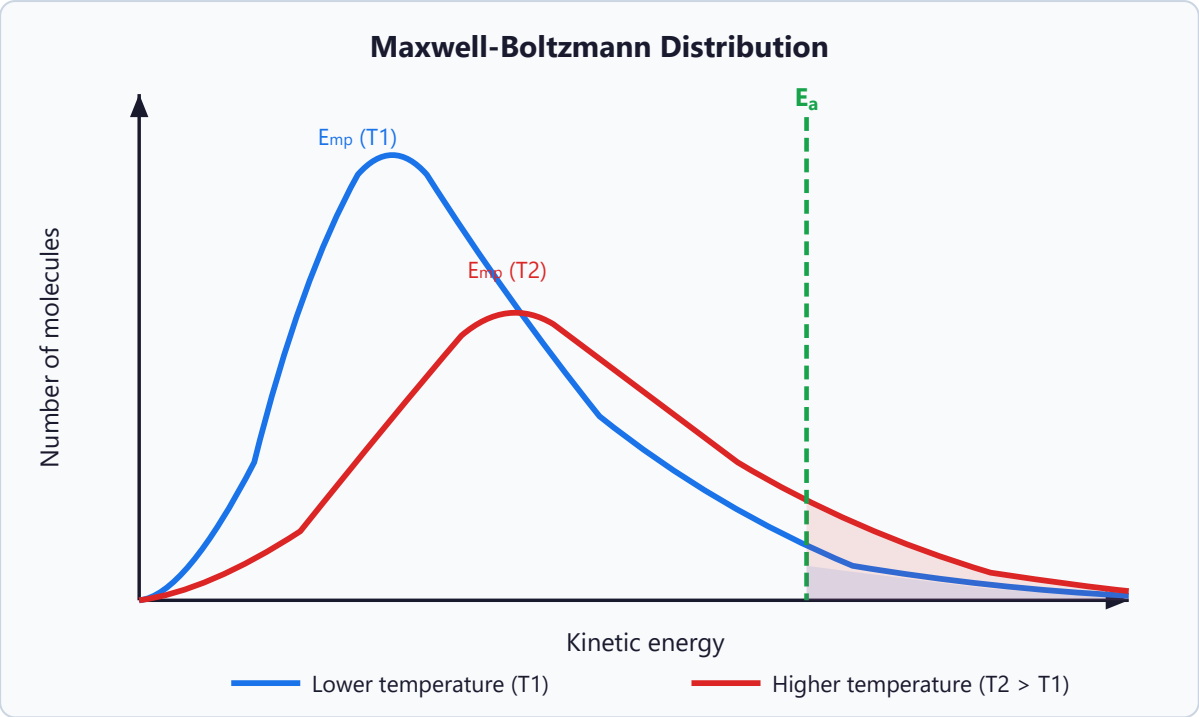
At a **higher temperature**:

- The peak shifts to the right (higher most probable energy) and becomes lower (flatter)
- The curve broadens — wider spread of energies
- The total area remains the same (same number of molecules)
- The area to the right of E_a increases significantly — many more molecules now have energy ≥ E_a
- This explains why a small temperature increase can cause a large increase in rate

Effect of a Catalyst on the Distribution

A catalyst does **not** change the shape or position of the Maxwell-Boltzmann curve (the energy distribution of molecules is unchanged). Instead, it **lowers E_a**, so the E_a line moves to the left on the diagram. This means a greater proportion of molecules now have energy ≥ the new, lower E_a, so the rate increases without changing the temperature.

Exam Tip: When drawing Maxwell-Boltzmann curves: (1) never let the curve touch the y-axis at the origin — the curve starts from exactly the origin point, (2) never let the curve cross or touch the x-axis at high energy — it approaches asymptotically, (3) always label axes correctly: "Number of molecules with energy E" (y-axis) vs "Kinetic energy" (x-axis), (4) at higher T, the peak must be lower and to the right, and both curves must enclose the same area.



2.4 Rate Equations and Orders of Reaction

Rate equation: A mathematical expression that relates the rate of reaction to the concentrations of reactants raised to their respective orders. For a reaction A + B → products:

Rate = $k[A]^m[B]^n$

where k = rate constant, m = order with respect to A, n = order with respect to B, and the **overall order** = m + n.

Key principles:

- Orders can **only be determined experimentally** — they CANNOT be deduced from the stoichiometric equation.
- Common orders are 0, 1, and 2 (they can be fractional but this is not covered at A-level).
- A species not in the rate equation has order 0 with respect to that species.
- The rate equation only includes species that appear in or before the rate-determining step.

Order	Effect of Doubling [A]	[A]-Time Graph Shape	Rate-[A] Graph Shape	Half-life Behaviour

0	No effect on rate	Linear decrease (straight line)	Horizontal line (rate = k)	Decreases with time
1	Rate doubles (×2)	Exponential decay curve	Straight line through origin	Constant (characteristic of 1st order)
2	Rate quadruples (×4)	Steeper curve than 1st order	Upward curve (parabola through origin)	Increases with time

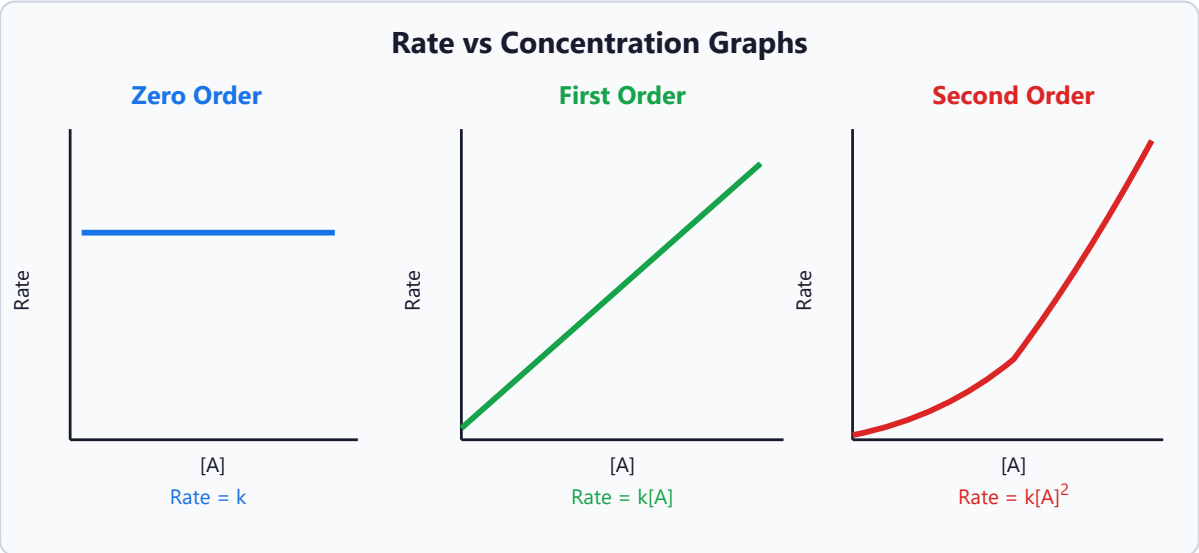
Determining Orders from Experimental Data

Method 1 — Initial rates method: Compare pairs of experiments where one reactant concentration changes while others are held constant.

- If doubling [A] doubles the rate: order = 1 (rate ∝ [A])
- If doubling [A] quadruples the rate: order = 2 (rate ∝ $[A]^2$)
- If doubling [A] has no effect: order = 0 (rate is independent of [A])
- If tripling [A] triples the rate: order = 1. If tripling [A] increases rate by 9×: order = 2.

Method 2 — Concentration-time graphs: For first-order reactions, successive half-lives are constant. Plot [A] vs time and measure the time for [A] to halve, then halve again, then halve again. If the half-life is constant, the reaction is first order.

Method 3 — Rate-concentration graphs: Plot initial rate vs [A]. Straight line through origin = first order. Curve (parabola) = second order. Horizontal line = zero order.



The Rate Constant, k

The rate constant k is a proportionality constant that is specific to a particular reaction at a particular temperature. Its value increases with temperature.

Units of k depend on the overall order of the reaction:

Overall Order	Units of k	How to derive
0	mol dm^{-3} s^{-1}	k = rate
1	s^{-1}	k = rate / [A]
2	mol^{-1} dm^3 s^{-1}	k = rate / $[A]^2$
3	mol^{-2} dm^6 s^{-1}	k = rate / $[A]^3$

Exam Tip: To work out units of k: rearrange the rate equation for k, substitute units for rate (mol dm^{-3} s^{-1}) and concentration (mol dm^{-3}), and simplify. This is a very common calculation question. Also remember that k changes with temperature but NOT with concentration.

2.5 The Arrhenius Equation

$$k = Ae^{-E_a/RT}$$

where A = pre-exponential factor (frequency factor, related to collision frequency and orientation), E_a = activation energy (J mol^{-1}), R = gas constant (8.314 J K^{-1} mol^{-1}), T = temperature (K).

The equation shows that k (and hence rate) increases exponentially with temperature. Taking natural logarithms gives the linear form:

$$\ln k = \ln A - E_a/RT$$

Plotting **ln k** (y-axis) against **1/T** (x-axis) gives a straight line with:

- **Gradient** = $-E_a/R$ (so E_a = -gradient × R)
- **y-intercept** = ln A

Sample calculation: If the gradient of an ln k vs 1/T plot is -5200 K, then $E_a = -(-5200) \times 8.314 = 43{,}200$ J mol^{-1} = 43.2 kJ mol^{-1}.

Exam Tip: T must be in Kelvin (K = °C + 273). When calculating the gradient, use two widely-spaced points on the line for accuracy. If given two values of k at two temperatures, you can calculate E_a using: $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$.

2.6 Reaction Mechanisms and the Rate-Determining Step

Most reactions occur in a series of elementary steps rather than in a single step. The overall rate of reaction is controlled by the **slowest step**, known as the rate-determining step (RDS).

Rate-determining step (RDS): The slowest step in a multi-step reaction mechanism. It acts as a "bottleneck" that controls the overall rate. The molecularity of the RDS determines the rate equation.

Intermediate: A species that is formed in one step and consumed in a subsequent step. It does not appear in the overall equation.

Connecting the rate equation to the mechanism:

- Only species involved in the RDS (or in equilibrium steps before the RDS) appear in the rate equation.
- If Rate = k[A][B], then the RDS involves one molecule of A and one of B (bimolecular).
- If Rate = $k[A]^2$, the RDS involves two molecules of A.
- If Rate = k[A] (and B is also a reactant), then A appears in the RDS but B does not — B must react in a fast step after the RDS.

Example: For the reaction $2NO_2 + F_2 \rightarrow 2NO_2F$, the rate equation is found experimentally to be Rate = $k[NO_2][F_2]$. This is consistent with the mechanism:

$$\text{Step 1 (slow, RDS): } NO_2 + F_2 \rightarrow NO_2F + F\bullet$$

$$\text{Step 2 (fast): } NO_2 + F\bullet \rightarrow NO_2F$$

The RDS involves one NO_2 and one F_2, matching the rate equation. F• is an intermediate.

Exam Tip: If asked to propose a mechanism consistent with a rate equation: (1) identify the species in the rate equation — these must be in the RDS, (2) the orders tell you how many of each species are in the RDS, (3) any remaining reactants from the overall equation must react in fast subsequent steps. Always check that your mechanism adds up to the overall equation.

2.7 Dynamic Equilibrium

Dynamic equilibrium: A state in a reversible reaction (in a closed system) where the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products remain constant (though not necessarily equal). Both forward and reverse reactions continue to occur — the system is dynamic, not static.

Conditions for dynamic equilibrium:

- The system must be **closed** (no matter enters or leaves)
- The reaction must be **reversible**
- The rates of forward and reverse reactions must be **equal**
- Macroscopic properties (concentration, colour, pressure) remain **constant**

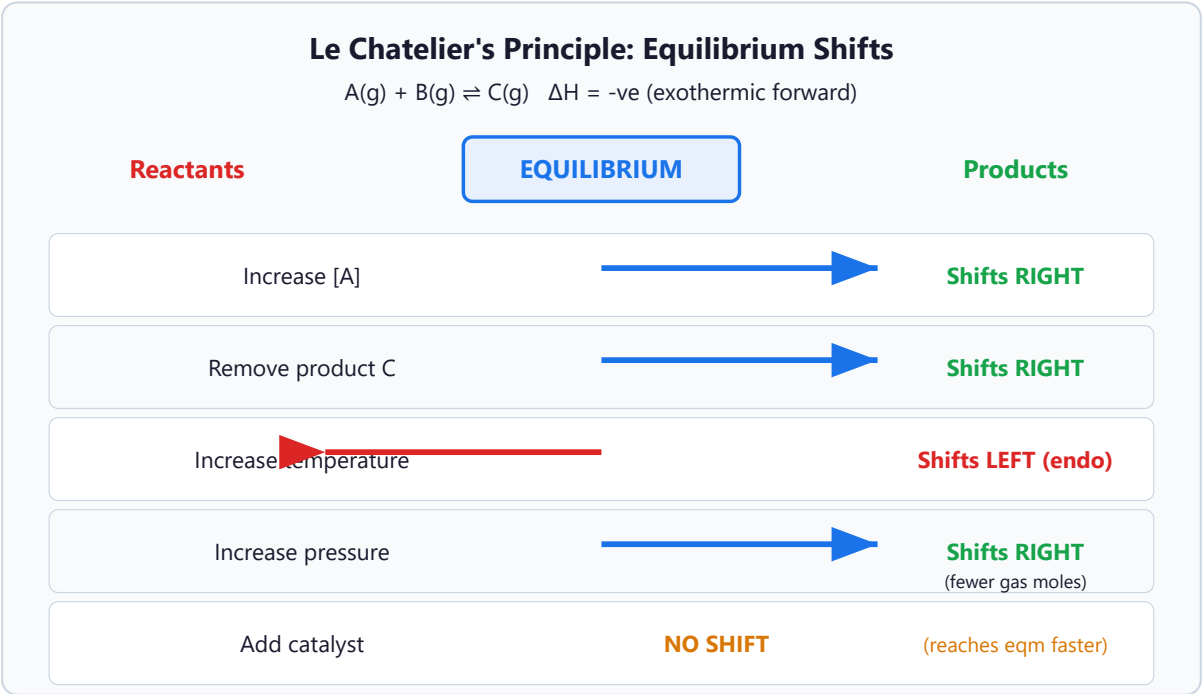
2.8 Le Chatelier's Principle

Le Chatelier's Principle: If a system at equilibrium is subjected to a change in conditions (concentration, pressure, or temperature), the position of equilibrium shifts in the direction that tends to oppose the change, establishing a new equilibrium.

Change Imposed	Effect on Equilibrium Position	Effect on K

Increase concentration of a reactant	Shifts right (towards products) to use up the added reactant	No change
Decrease concentration of a product (remove it)	Shifts right to replace the removed product	No change
Increase temperature (exothermic forward rxn)	Shifts left (endothermic direction) to absorb excess heat	K decreases
Increase temperature (endothermic forward rxn)	Shifts right (endothermic direction)	K increases
Increase pressure (for gaseous equilibria)	Shifts towards the side with fewer moles of gas	No change
Add a catalyst	No shift — equilibrium reached faster, but position unchanged	No change

Critical point: Only temperature changes affect the value of the equilibrium constant K. Changes in concentration, pressure, and addition of a catalyst alter the position of equilibrium (the relative amounts of products and reactants) but NOT the value of K.



2.9 Equilibrium Constants: K_c and K_p

K_c (Concentration Equilibrium Constant)

For the equilibrium aA + bB ⇌ cC + dD:

$$K_c = [C]^c[D]^d / [A]^a[B]^b$$

All concentrations are **equilibrium concentrations** in mol dm^{-3}. Pure solids and pure liquids are NOT included in the K_c expression (their concentrations are constant). Homogeneous equilibria have all species in the same phase; heterogeneous equilibria have species in different phases.

Units of K_c: Depend on the stoichiometry. Work them out by substituting (mol dm^{-3}) for each concentration term and simplifying. If total moles of products = total moles of reactants, K_c has no units.

Interpreting K_c:

- $K_c >> 1$: Equilibrium lies far to the right (products favoured)
- $K_c << 1$: Equilibrium lies far to the left (reactants favoured)
- $K_c \approx 1$: Significant amounts of both reactants and products at equilibrium

K_p (Pressure Equilibrium Constant)

For gaseous equilibria, K_p uses partial pressures instead of concentrations:

$$K_p = p(C)^c \times p(D)^d / p(A)^a \times p(B)^b$$

The partial pressure of a gas is calculated from its mole fraction:

$$\text{Mole fraction of A} = \text{moles of A at equilibrium} / \text{total moles of gas at equilibrium}$$

$$p(A) = \text{mole fraction of A} \times \text{total pressure}$$

Key relationship: The sum of all mole fractions = 1, and the sum of all partial pressures = total pressure.

Exam Tip: For K_p calculations, you often need to construct an ICE table (Initial, Change, Equilibrium) to find equilibrium moles, then calculate mole fractions, then partial pressures, then K_p. Show your working clearly and include units. Remember: pure solids are excluded from K_p expressions too (e.g. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ gives $K_p = p(CO_2)$).

2.10 Oxidation States and Redox

Oxidation: Loss of electrons / increase in oxidation state. **OIL** — Oxidation Is Loss
Reduction: Gain of electrons / decrease in oxidation state. **RIG** — Reduction Is Gain

Oxidising agent (oxidant): A species that accepts electrons from another species (it is itself reduced).
Reducing agent (reductant): A species that donates electrons to another species (it is itself oxidised).

Rules for Assigning Oxidation States

1. Elements in their standard state: oxidation state = **0** (e.g. O_2, Fe, Na, Cl_2)
2. Monatomic ions: oxidation state = **ion charge** (e.g. Na^+ = +1, Cl^- = -1, Fe^{3+} = +3)
3. Oxygen: usually **-2** (exceptions: peroxides like H_2O_2 = -1, OF_2 = +2)
4. Hydrogen: usually **+1** (exception: metal hydrides like NaH = -1)
5. Fluorine: always **-1** (most electronegative element)
6. Sum of oxidation states in a neutral compound = **0**
7. Sum of oxidation states in a polyatomic ion = **ion charge**

Disproportionation: A reaction in which the same element is simultaneously oxidised and reduced. Example: $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$. Chlorine starts at oxidation state 0 and is both reduced to -1 (in NaCl) and oxidised to +1 (in NaClO).

2.11 Redox Equations and Half-Equations

Half-equations show oxidation and reduction separately. They are useful for balancing complex redox equations.

Steps to balance half-equations in acidic solution:

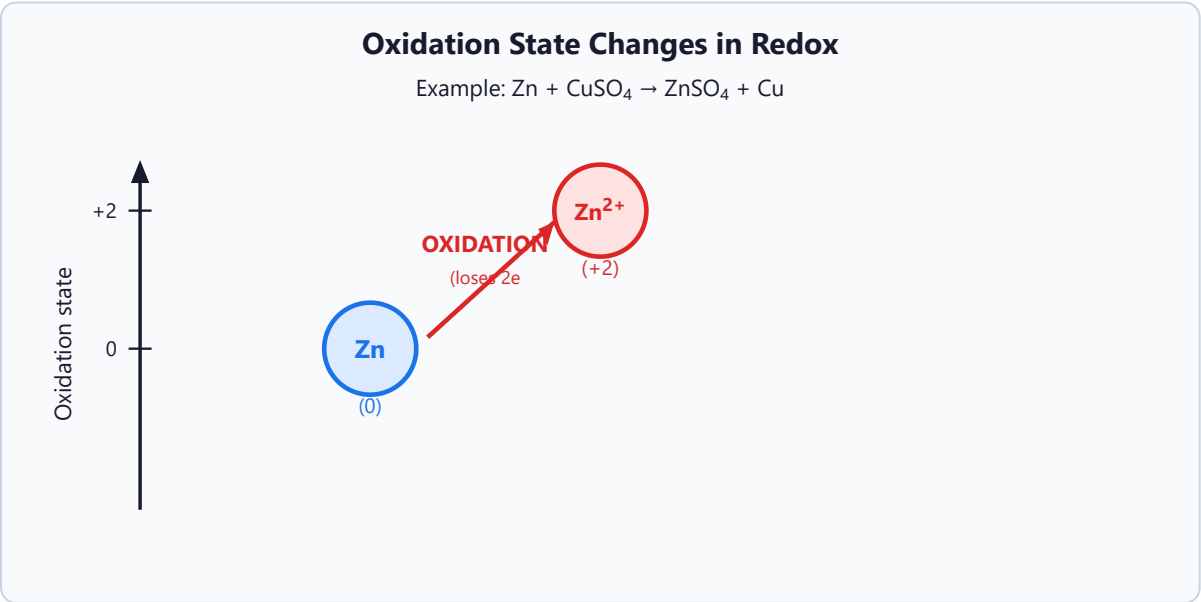
1. Write the unbalanced half-equation with the species being oxidised or reduced
2. Balance all atoms except O and H
3. Balance O by adding H_2O to the side that needs oxygen
4. Balance H by adding H^+ to the side that needs hydrogen
5. Balance charge by adding electrons (e^-) to the more positive side
6. To combine half-equations: multiply so the number of electrons in both half-equations is equal, then add them together (electrons should cancel)

Example — MnO_4^- reacting with Fe^{2+}:

$$\text{Reduction: } MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$\text{Oxidation: } Fe^{2+} \rightarrow Fe^{3+} + e^- \quad (\times 5)$$

$$\text{Overall: } MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$



⁻, OS increases) Cu2+ (+2) Cu (0) REDUCTION (gains 2e⁻, OS decreases) OIL (Oxidation Is Loss of e⁻) RIG (Reduction Is Gain of

e^-)

Exam Tip: Always check that atoms AND charges balance on both sides of the final equation. Electrons must cancel completely — they should NOT appear in the overall balanced equation. When asked to identify the oxidising/reducing agent, remember: the oxidising agent is REDUCED (it gains electrons), and the reducing agent is OXIDISED (it loses electrons). This causes frequent confusion.

2.12 Electrochemical Cells and Electrode Potentials

Electrochemical cell: A device that converts chemical energy into electrical energy (galvanic/voltaic cell) or uses electrical energy to drive a non-spontaneous reaction (electrolytic cell).

Standard electrode potential (E°): The EMF of a half-cell measured relative to the standard hydrogen electrode (SHE) under standard conditions (298 K, 100 kPa, 1.00 mol dm^{-3} solutions). The SHE is defined as E° = 0.00 V.

A standard electrochemical cell consists of two half-cells connected by:

- A **salt bridge** (e.g. filter paper soaked in KNO_3) — allows ion flow to complete the internal circuit and maintain electrical neutrality in each half-cell
- An **external wire** — allows electron flow from anode to cathode
- A **high-resistance voltmeter** — measures EMF without drawing significant current

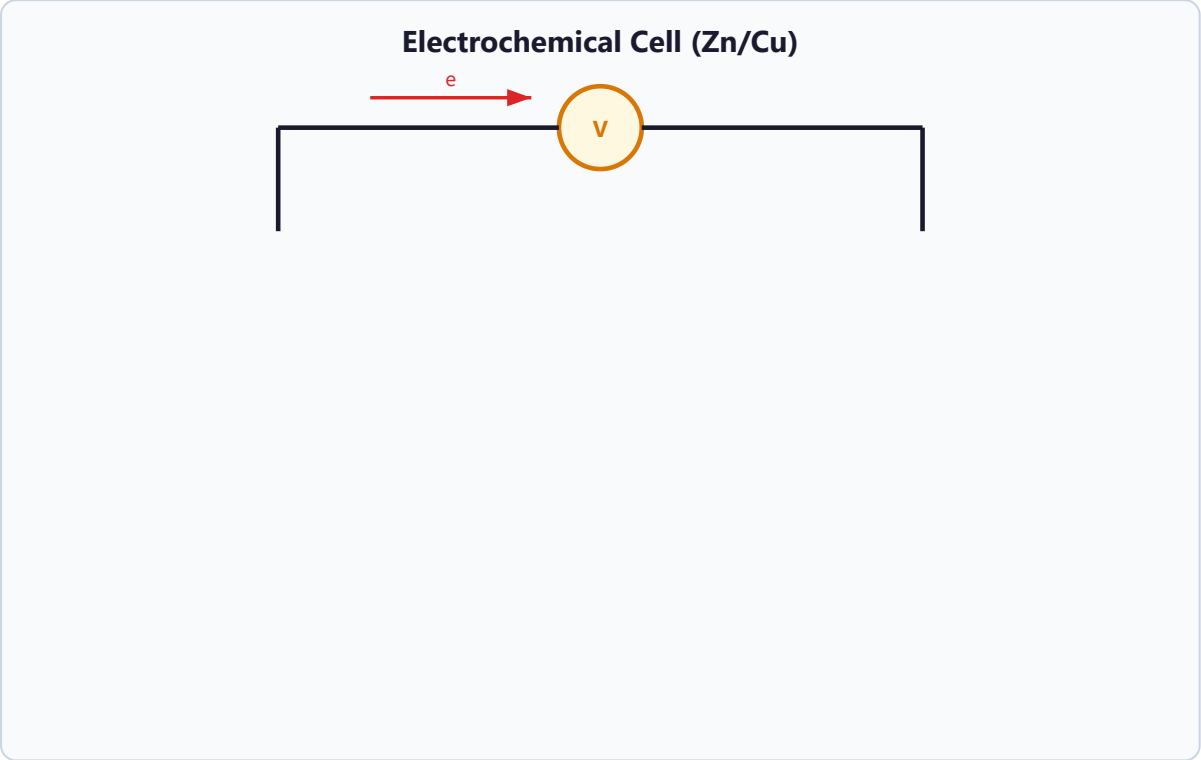
Calculating Cell EMF

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{most\ positive} - E^\circ_{most\ negative}$$

Key rules:

- The half-cell with the more positive E° is the cathode (reduction occurs here)
- The half-cell with the more negative E° is the anode (oxidation occurs here)
- Electrons flow from anode (−) to cathode (+) through the external circuit
- Conventional current flows in the opposite direction to electron flow

- A **positive $E°_{cell}$** indicates a thermodynamically feasible (spontaneous) reaction



- flow Zn(s) ANODE (-) Oxidation ZnSO4(aq) 1.0 mol dm-3 Cu(s) CATHODE (+) Reduction CuSO4(aq) 1.0 mol dm-3 Salt bridge (KNO3) Zn → Zn2+ + 2e- Cu2+ + 2e- → Cu E°cell = (+0.34) - (-0.76) = +1.10 V

Predicting Feasibility Using E° Values

A reaction is thermodynamically feasible if $E°_{cell} > 0$. To predict whether a reaction will occur:

1. Identify the two relevant half-equations from the data booklet.
2. The half-cell with the more positive E° undergoes reduction (gains electrons).
3. The half-cell with the more negative E° undergoes oxidation (loses electrons) — reverse this half-equation.
4. Calculate $E°_{cell} = E°_{cathode} - E°_{anode}$. If positive, the reaction is feasible.

Limitations of using E° values to predict feasibility:

- **Kinetics:** A positive $E°_{cell}$ tells you the reaction is thermodynamically favourable but says nothing about rate. The reaction may have a very high activation energy and be extremely slow (e.g. the thermite reaction Fe_2O_3 + Al is feasible but requires ignition).

- **Non-standard conditions:** E° values apply only at standard conditions. Non-standard concentrations, temperatures, or pressures will change the actual cell potential (described by the Nernst equation at university level).
- **Overpotential:** In electrolysis, extra voltage is needed to overcome kinetic barriers at the electrode surface.

The Electrochemical Series

Half-equations are listed in order of E° from most negative (top, strongest reducing agents) to most positive (bottom, strongest oxidising agents). A species on the right of a half-equation can oxidise any species on the left of a half-equation above it in the series.

Exam Tip: When using electrode potential data: (1) "more positive E° = stronger oxidising agent" (better at gaining electrons), (2) "more negative E° = stronger reducing agent" (better at losing electrons), (3) never change the E° value when reversing a half-equation — E° is an intensive property, (4) never multiply E° values when balancing half-equations, (5) always state whether a reaction is feasible AND comment on potential kinetic limitations.

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